

QUICK START



Tankie

AI Tank Battle — get your bot running in minutes

1 What is Tankie?

Tankie is a **turn-based 2-player tank battle** played on a procedurally generated maze. You don't control the tank yourself — you write an AI strategy that plays for you. The server runs the game; your client just sends actions each turn.

2 Prerequisites

You need	Version	Purpose
Godot	4.6	Game server (provided)
Python or C# or C++	3.11+ / .NET 8 / C++17	Your client language

3 Start the server

Open the `tankie/` folder in Godot 4.6 and press **F5**. The WebSocket server starts automatically on `ws://localhost:8080/`.

```
# or from the command line:
cd tankie
godot           # with game window (recommended)
godot --headless # no window
```

4 Run your client

Pick one language skeleton, set your tank name, and run it:

Python

```
cd python-skeleton
pip install websockets
python client.py
```

C#

```
cd csharp-skeleton/TankieClient
dotnet run
```

C++

```
cd cpp-skeleton
cmake -B build && cmake --build build
./build/tankie_client
```

Set your tank's name (ID) in the entry file: `TANK_ID` in `client.py` / `TankId` in `Program.cs` / `TANK_ID` in `main.cpp`.

5 Write your strategy

The only file you need to edit is the **strategy file** for your language. It receives a snapshot of the game state and must return a list of actions.

```
# python-skeleton/strategy.py
def decide_actions(state):
    me = state.my_tank()          # my position & turret angle
    enemies = state.opponents()  # list of opponent tanks

    # Build your action list and return it:
    return [{"type": "move", "direction": "up"}]
```

Tip: You can return up to 2 moves, or 1 move + 1 shoot, or just 1 shoot. Rotations are free and don't count against your budget.

6 Start the game

01

Both clients connect

02

Press Start

03

Play!

- Both players must be in the lobby.
- Click Start in the Godot game window.
- Turns alternate. Last tank standing wins.

Note: An over-budget action list is silently rejected — the server skips your entire turn. Stay within budget!

7 Action budget cheat-sheet

Combination	Allowed?
Move + Move	✓ Yes
Move + Shoot	✓ Yes
Shoot only	✓ Yes
Move only	✓ Yes
Rotate (any number)	✓ Always free
Move + Move + Shoot	✗ Over budget
Shoot + Shoot	✗ Over budget